

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\`a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-adf2141228d35b940.jsonl`  
- SHA-256: `8baf263e85c1e4440e2db11e183e2b55e83ddea0b126d2d41ba82d9574aa5a46`  
- Source modified: `2026-05-31T04:26:19+00:00`  
- Imported at: `2026-07-05T16:48:21+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:26:01.613Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"The benchmark evaluates table recognition with a dedicated Table TEDS metric, providing an objective, comparable accuracy measure across both OCR/pipeline tools and VLMs specifically for table-structure reconstruction ? exactly the capability muneem needs for dense, semi-ruled statement tables."

****Source:**** <https://github.com/opendatalab/OmniDocBench> (primary)

****Supporting quote:**** "Both approaches are evaluated on identical metrics (Edit Distance, TEDS, CDM for formulas)."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:26:04.614Z)

I'll verify this claim about OmniDocBench's TEDS metric for table recognition. Let me start by examining the primary source and searching for corroborating/contradicting evidence.

3. assistant (2026-05-31T04:26:07.499Z)

Now let me fetch the OmniDocBench primary source and search for the actual benchmark structure.

4. user (2026-05-31T04:26:15.061Z)

OmniDocBench: Table Recognition & Metrics Analysis

Table Recognition Evaluation

Yes, OmniDocBench comprehensively evaluates table recognition. The benchmark includes "bounding box information for tables on each PDF page along with corresponding table recognition annotations."

Table-Specific Metrics

- For tables, the framework employs:
- **TEDS** (Tree Edit Distance based Similarity) - measures structural similarity
 - **Edit Distance** - captures character-level accuracy
 - **Table TEDS-S** - a variant providing supplementary structural assessment

Unified Evaluation Across Model Types

The benchmark evaluates both specialized OCR systems and general vision-language models using identical metrics. The leaderboard includes:

- **Specialized VLMs**: MinerU2.5, GLM-OCR, PaddleOCR-VL
- **General VLMs**: Gemini 3 Pro, GPT-4o, Qwen3-VL
- **Pipeline Tools**: Marker, Mathpix

All receive evaluation on the same metric suite.

Complete Metrics by Task Category

Category	Metrics
Text blocks	Edit Distance, BLEU, METEOR
Display formulas	Edit Distance, CDM
Tables	TEDS, Edit Distance
Reading order	Edit Distance
Layout detection	COCODet (mAP, mAR)

This unified framework ensures fair comparison across diverse model architectures.

5. user (2026-05-31T04:26:19.444Z)

Web search results for query: "OmniDocBench TEDS table recognition metric VLM pipeline evaluation"

Links: [{"title": "OmniDocBench: Benchmarking Diverse PDF Document Parsing with Comprehensive Annotations", "url": "https://arxiv.org/html/2412.07626v1"}, {"title": "OmniDocBench v1.0 Benchmark", "url": "https://www.emergentmind.com/topics/omnidocbench-v1-0"}, {"title": "OmniDocBench: A Real-World, Fine-Grained Benchmark for Fair and Comprehensive PDF Document Parsing Evaluation | PaperCodex", "url": "https://www.papercodex.com/omnidocbench-a-real-world-fine-grained-benchmark-for-fair-and-comprehensive-pdf-document-parsing-evaluation/"}, {"title": "OmniDocBench is Saturated, What's Next for OCR Benchmarks?", "url": "https://www.llamaindex.ai/blog/omnidocbench-is-saturated-what-s-next-for-ocr-benchmarks"}, {"title": "OmniDocBench: Benchmarking Diverse PDF Document Parsing with", "url": "https://openaccess.thecvf.com/content/CVPR2025/papers/Ouyang_OmniDocBench_Benchmarking_Diverse_PDF_Document_Parsing_with_Comprehensive_Annotations_CVPR_2025_paper.pdf"}, {"title": "1 Performance of GLM-OCR on OmniDocBench v1.5.", "url": "https://arxiv.org/html/2603.10910v1"}, {"title": "OmniDocBench v1.5 Benchmark", "url": "https://www.emergentmind.com/topics/omnidocbench-v1-5"}, {"title": "OmniDocBench Leaderboard: PDF Document Parsing SOTA | CodeSOTA", "url": "https://www.codesota.com/browse/computer-vision/document-parsing/omnidocbench"}]

Based on the search results, here's what I found about OmniDocBench, TEDS table recognition metrics, and VLM pipeline evaluation:

OmniDocBench Overview

OmniDocBench v1.0 is a comprehensive benchmark evaluating document parsing systems on diverse real-world PDFs with detailed annotations and multi-level metrics, designed to evaluate document parsing systems with extensive high-quality annotations across varied document types.

TEDS Metric for Table Recognition

Table recognition evaluation uses TEDS (Tree-Edit-Distance-based Similarity) and Edit Distance on HTML/LaTeX outputs, with automatic LaTeX-to-HTML conversion via LaTeXML. The evaluation pipeline uses Normalized Edit Distance for text, Tree-Edit-Distance-based similarity (TEDS) for tables, and Character Detection Matching (CDM) for formulas.

VLM Pipeline Evaluation Results

The benchmark evaluates different types of models:

- **OCR-based models**: OCR-based models demonstrate superior overall performance, with RapidTable achieving the highest scores in language diversity and maintaining stable performance across different frame types.

- **General VLMs**: General VLMs score ~71.74 TEDS on table recognition, while OCR-based RapidTable achieves ~82.5 TEDS.

- **Specialized VLMs**: Expert VLMs show competitive results in specific scenarios, with StructEqTable excelling in no frame tables and showing better rotation robustness.

End-to-end assessments of mainstream document parsing methods evaluate their performance from input PDF images to the resultant document parsing outputs.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:26:19.818Z)

You've hit your session limit · resets 1:30pm (Asia/Calcutta)